```
!pip uninstall -y langchain
!pip install langchain langchain-community
```

```
Found existing installation: langchain 1.2.14
Uninstalling langchain-1.2.14:
  Successfully uninstalled langchain-1.2.14
Collecting langchain
  Downloading langchain-1.2.15-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: langchain-community in /usr/local/lib/python3.12/dist-packages (0.4.1)
Requirement already satisfied: langchain-core<2.0.0,>=1.2.10 in /usr/local/lib/python3.12/dist-packages (from langchain) (1.2.
Collecting langgraph<1.2.0,>=1.1.5 (from langchain)
  Downloading langgraph-1.1.6-py3-none-any.whl.metadata (8.0 kB)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.12.3)
Requirement already satisfied: langchain-classic<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain-comm
Requirement already satisfied: SQLAlchemy<3.0.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community)
Requirement already satisfied: requests<3.0.0,>=2.32.5 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (
Requirement already satisfied: PyYAML<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (6.0
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (3.
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-comm
Requirement already satisfied: dataclasses-json<0.7.0,>=0.6.7 in /usr/local/lib/python3.12/dist-packages (from langchain-commu
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from langchain-com
Requirement already satisfied: langsmith<1.0.0,>=0.1.125 in /usr/local/lib/python3.12/dist-packages (from langchain-community)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (
Requirement already satisfied: numpy>=1.26.2 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (2.0.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3-
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langch
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langc
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->lan
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langch
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langc
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.
Requirement already satisfied: typing-inspect<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7.
Requirement already satisfied: langchain-text-splitters<2.0.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from langcha
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.1
```

```
df = pd.read_csv("/mnt/data/rag_dataset_lab21 (1).csv")

# Convert dataset into text documents
documents = df.astype(str).apply(lambda x: " ".join(x), axis=1).tolist()

print(documents[:5])
```

```
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->lang
```

```
embedding_model = HuggingFaceEmbeddings(model_name="all-MiniLM-L6-v2")
```

```
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4-
```

```
db = Chroma.from_texts(documents, embedding_model)
```

```
Requirement already satisfied: python-dotenv>=0.21.0 in /usr/local/lib/python3.12/dist-packages (from pydantic-settings<3.0.0,
```

```
retriever = db.as_retriever()
```

```
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->lang
```

```
os.environ["OPENAI_API_KEY"] = "your_api_key_here"

llm = OpenAI()
```

```
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->lan
```

```
qa_chain = RetrievalQA.from_chain_type(
    llm=llm,
    retriever=retriever
)
```

```
Downloading langgraph-1.1.6-py3-none-any.whl (169 kB)
```

```
query = "What is Artificial Intelligence?"
response = qa_chain.run(query)

print(response)
```

```
docs = retriever.get_relevant_documents(query)
```